

Regression Assignment DOC

1) PROBLEM IDENTIFICATION:

a) *Domain Selection:*

- i) Machine Learning

b) *Learning Selection:*

- i) Supervised learning (requirements are very clear
And I/P and O/P well defined)

c) *Classification or Regression:*

- i) Regression (continues numerical value)

2) BASIC INFO ABOUT THE DATASET:

a) *Total no of rows* = **1338**

b) *Total no of columns* = **6**

3) PRE-PROCESSING METHODS:

In this dataset two column has categorical data they are 'SEX' and 'SMOKER' so in AI algorithm can't handle these type of data, so that I converted as numerical data by using 'ONE HOT ENCODING' in 'NOMINAL' type.

4) R2_SCORE OF THE MODELS:

1. *Multiple Linear Regression:*

R2 value = 0.7238

2. Support Vector Machine:

S.NO	Hyper Tuning Parameter	Linear (r value)	RBF (Non Linear) (r value)	Poly (r value)	Sigmoid (r value)
1	C=10	-1.217	-404.66	-121.869	-84.603
2	C=100	0.047	-3.826	-0.100	-0.726
3	C=500	0.748	0.297	0.778	-0.406
4	C=1000	0.771	0.733	0.833	-0.068
5	C=2000	0.771	0.816	0.846	-0.097
6	C=3000	0.771	0.845	0.846	-0.228

The SVM Regression use **R2 Value** (poly and hyper parameter (C=3000)) = **0.846**

3. Decision Tree:

S.NO	Criterion	Max Features	Splitter	R Value
1	squared_error	None	best	0.762
2	squared_error	None	random	0.714
3	squared_error	sqrt	best	0.703
4	squared_error	sqrt	random	0.678
5	squared_error	log2	best	0.759
6	squared_error	log2	random	0.676
7	absolute_error	None	best	0.714
8	absolute_error	None	random	0.763
9	absolute_error	sqrt	best	0.786

10	absolute_error	sqrt	random	0.722
11	absolute_error	log2	best	0.770
12	absolute_error	log2	random	0.752
13	poisson	None	best	0.792
14	poisson	None	random	0.779
15	poisson	sqrt	best	0.737
16	poisson	sqrt	random	0.741
17	poisson	log2	best	0.772
18	poisson	log2	random	0.712

The Decision Tree Regression use **R2 Value**
 (poisson,None,best) = **0.792**

4. Random Forest:

S.NO	Criterion	Max Features	Random_state	n_estimators	R Value
1	squared_error	None	None	10	0.855
2	squared_error	None	None	100	0.865
3	squared_error	sqrt	None	10	0.859
4	squared_error	sqrt	None	100	0.876
5	squared_error	log2	None	10	0.861
6	squared_error	log2	None	100	0.874
7	absolute_error	None	None	10	0.848
8	absolute_error	None	None	100	0.862
9	absolute_error	sqrt	None	10	0.851
10	absolute_error	sqrt	None	100	0.872
11	absolute_error	log2	None	10	0.850

12	absolute_error	log2	None	100	0.873
13	poisson	None	None	10	0.859
14	poisson	None	None	100	0.866
15	poisson	sqrt	None	10	0.853
16	poisson	sqrt	None	100	0.872
17	poisson	log2	None	10	0.870
18	poisson	log2	None	100	0.881

The Random Forest Regression use **R2 Value**
 (poisson,log2,100) = **0.881**

5) FINAL MODEL:

Compare to all other models RF performed as good.
 Move to deployment phase.

a) Random Forest Regression – R2 Value = 0.88